

(1) Demand and the law of demand.

⇒ Demand is the consumer's desire and ability to purchase goods and services.

* Law of demand:

→ Law of demand states that the demand is inversely proportional to the price of goods and services if other things remaining constant.

i.e. Demand $\propto \frac{1}{\text{Price}}$

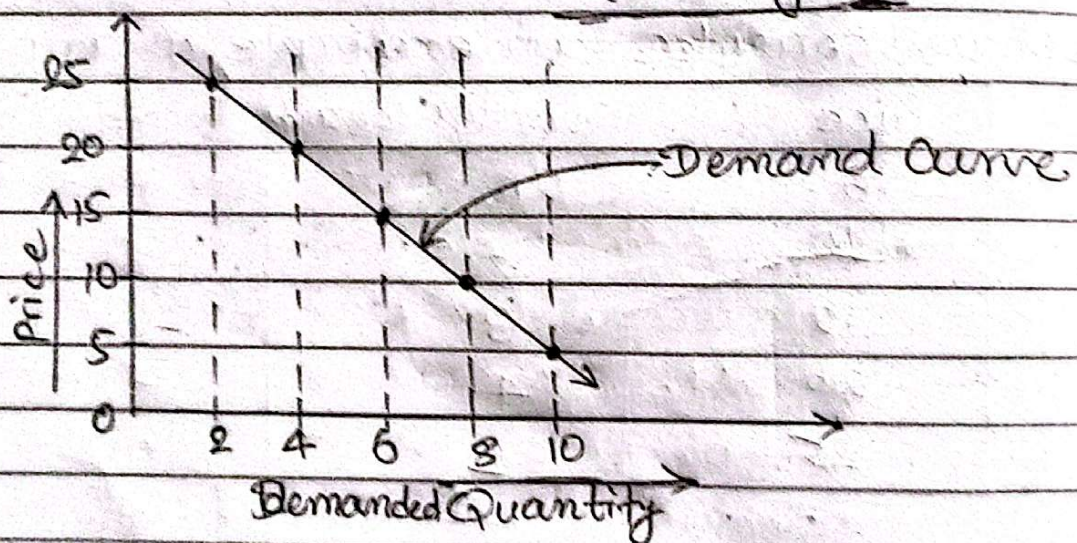
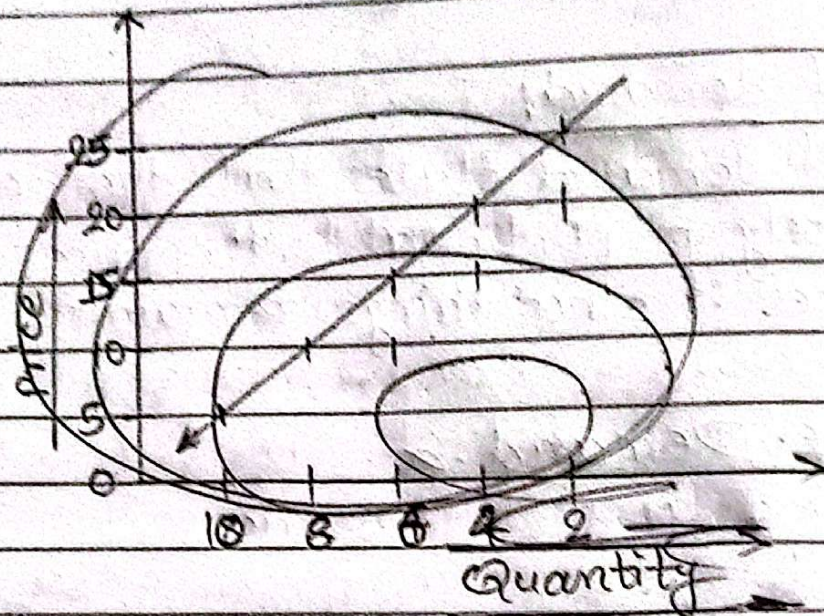
Let us consider an example of apple.

Price	Quantity Demanded
5	10
10	8
15	6
20	4
25	2

Fig. Demand schedule

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The above demand schedule is represented by following graph.



- From the above graph, it is clear that the demand curve is downward sloping.
- The higher the price of goods, the lower will be the demanded quantity and vice-versa.

(2) Supply and law of supply:

→

(2) Factors affecting Demand:

- Income of the people
- Taste of Consumers
- Prices of related goods
- Customs and fashions
- Weather
- Future expectations
- Price of commodity.

(3) Supply and Law of Supply:

→ Supply is the desire with ability to sell and willingness to sell.

* Law of Supply:

→ Law of supply states that the supply is directly proportional to price of goods and services if other things remaining constant.

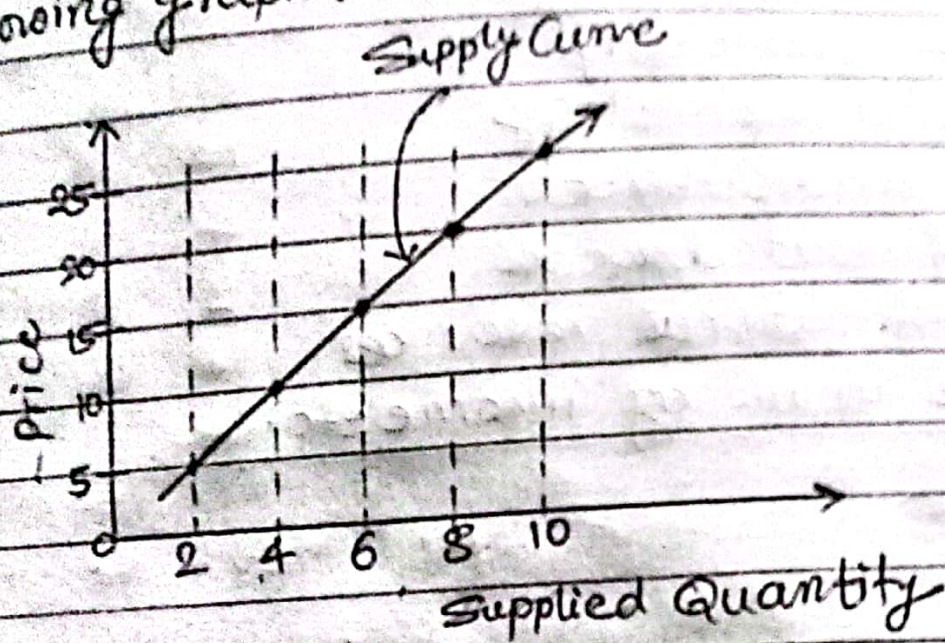
i.e. $\text{Supply} \propto \text{Price}$.

Let us consider an example of supply:

Price	Quantity Demand Supplied
5	2
10	4
15	6
20	8
25	10

Fig. supply schedule

The above supply schedule is represented in the following graph:



From the above graph, it is clear that the supply ~~curve~~ curve is upward sloping.

The higher the price of goods, the higher will be the supply and vice-versa.

①

(4) Factors Affecting Supply:

- New Inventions
- Price of commodity
- Production technology
- Taxes and subsidies
- Price of related goods
- State of natural resources
- Change in money income etc.

(5) Law of Diminishing Marginal Utility:

→ The law of diminishing marginal utility states that the marginal utility gradually decreases with the level of consumption utility being defined as benefit or satisfaction.

→ This law is based and explained under certain assumptions:

- (i) Utility is measurable
- (ii) Consumers satisfaction
- (iii) Commodity is measurable
- (iv) Consumers taste and preferences unchanged.
etc.

→ Suppose a person eats ~~an~~ apple, and 1st unit of apple gives him maximum satisfaction.

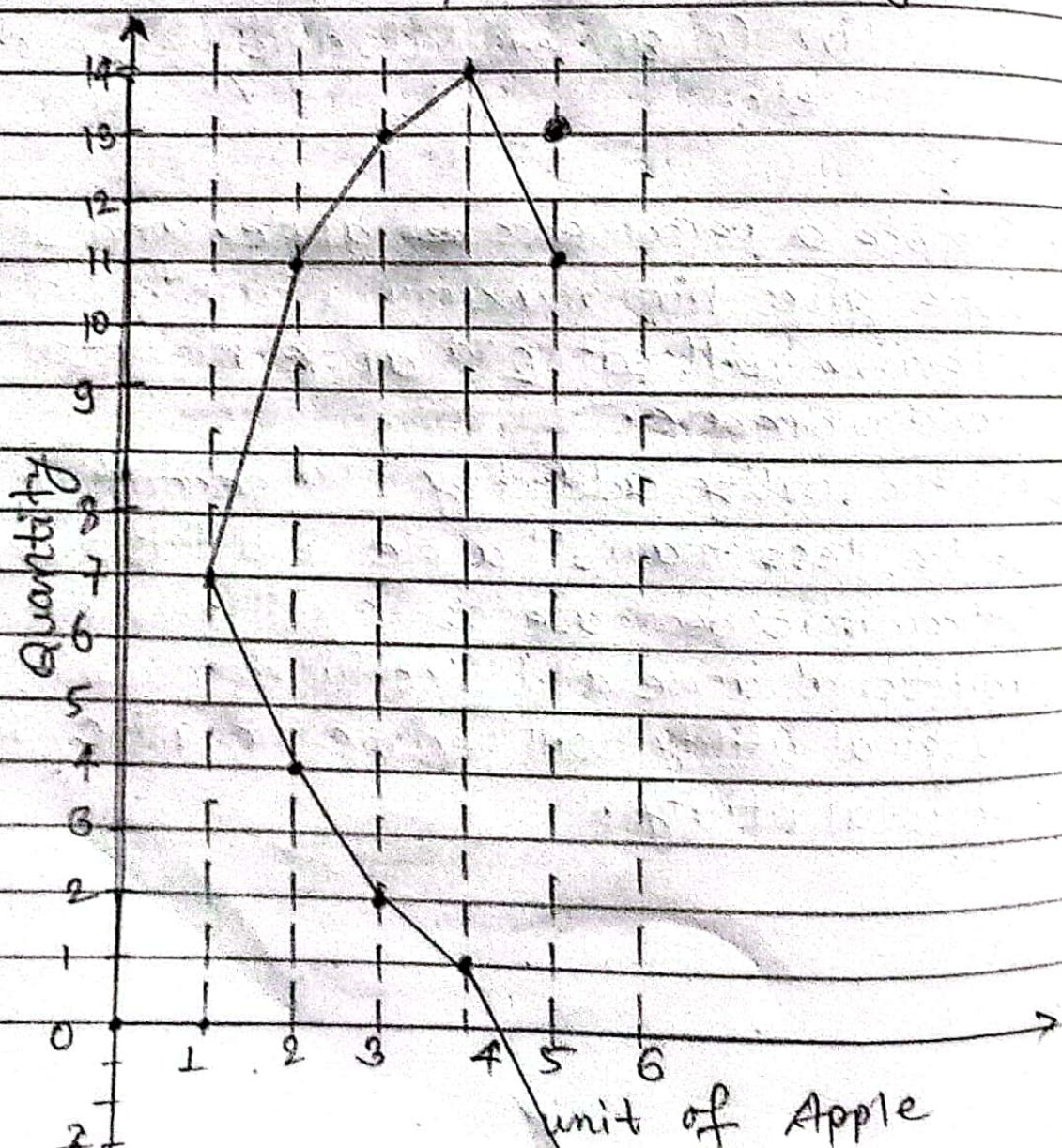
→ When he will eat 2nd apple, his total satisfaction would increase.

→ But the utility added by 2nd second apple will be less than 1st apple and this marginal utility will continue decrease as he will keep on eating additional apple and once his stomach is full, the marginal utility will become negative, decreasing the total utility.

→ Let us consider the following table,

Apple	Total utility (TU)	Marginal Utility (MU)
0	0	0
1	7	7
2	11	4
3	13	2
4	14	1
5	11	-3

The above table is represented in the graph as:



(6) "Knowledge of engineering economics helps in decision making process." Justify it by the principle of engineering economics.

Engineering economics is the science that deals with decision-making methods towards maximizing costs and maximizing benefits to the business organization.

The principles of engineering economics are as follows:

(i) Develop the alternatives:

→ The fundamental step in decision making is to identify and develop a feasible alternatives.
(define)

(ii) Focus on differences:

→ When comparing alternatives, it is crucial to focus on the differences in their expected outcomes such as cost, benefits and risks.

(iii) Use a consistent viewpoint:

→ All alternatives should be evaluated from a consistent viewpoint / perspective, ensuring that comparisons are fair and unbiased.

(iv) Use a common unit of measure;

→ To facilitate comparison, it is essential to measure costs and benefits using a common unit.

(v) Consider all relevant criteria;

→ Selection of alternatives requires the use of a preferred criteria. ~~such~~ For example; long term interest.

(vi) Make uncertainty explicit;

→ Risks and uncertainty are inherent in estimating the future outcomes of the alternatives.

(vii) Revisit your decisions;

→ After implementing a decision, it is important to monitor and be ready to revisit and adjust ~~the~~ the choices as needed.

The first two principles focuses on setting up the process, the next two three focuses on evaluation criteria and the last two focuses on analysis.

Thus, knowledge of engineering economics helps in decision making process.

→ Engineering economics is the science that deals with making-decisions methods towards minimizing costs and maximizing benefits to the business organization.

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(1) Manufacturing cost:

→ Cost refers to the amount of expenditure incurred in acquiring something.

(2) Manufacturing Costs:

→ Manufacturing costs are the costs of all resources consumed in the process of making a product.

→ Manufacturing costs are also known as Production costs.

→ Manufacturing costs are directly associated with the production of goods.

→ Manufacturing costs include direct labor costs, direct materials costs and manufacturing overhead costs.

(i) Direct materials costs:

→ These costs are the cost of raw materials that are used to produce the product.

→ For example: Woods for furniture, metals for machinery etc.

(ii) Direct labor costs:

→ These ~~are~~ costs are the wages of workers directly involved in production of product.

→ Examples: Welders in metal fabrication, carpenters in building works etc.

(iii) Manufacturing Overhead Costs:

→ These costs are indirect costs related to the production of product.

→ For example: Factory utilities, equipment depreciation etc.

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(3) Non-manufacturing costs;

- Non-manufacturing costs are the costs of company operations that are not directly related to the productions.
- Non-manufacturing costs are also known as period costs.
- Non-manufacturing costs are not directly associated with production of goods.
- Non-manufacturing costs include marketing costs, administrative costs, distribution costs, Research & Development costs.

(i) Marketing costs;

- Marketing costs are the costs related to securing the customers orders and delivering the product.
- Examples: Marketing Sales Commission etc.

(ii) Administrative costs;

- Administrative costs are the costs associated with the overall management.
- Examples: Office salaries, executive compensation etc.

(iii) Distribution costs

(iv) Research and Development costs

(4) Opportunity Costs:

- Opportunity costs are the benefits that you give up by choosing one alternative over another.
- Opportunity costs are relevant for future decisions.
- Opportunity costs reflect the potential value of the alternatives that you forgo.
- For example:
If you spend time studying instead of working, the money you could have earned is the opportunity cost.

(7) Differential Costs:

- Differential cost is the difference in costs between any two alternatives.
- It is also known as Incremental costs.

(f) Sunk costs:

- Sunk costs are the costs that you have already incurred and cannot recover, regardless of your future actions.
- Sunk costs are ~~not~~ not relevant for future decisions.
- Sunk costs reflect the past value of the alternatives that you choose.
- For example:
If you buy a movie ticket but don't watch the movie, the ticket cost is the sunk cost.

(g) Marginal Cost:

- Marginal Cost is the ratio of change in total cost to the change in quantity.
- i.e. $MC = \frac{\Delta TC}{\Delta Q}$

→ For example;

$$C_1 = 1000 ; C_2 = 2000$$

$$Q_1 = 100 , Q_2 = 200$$

$$\therefore MC = \frac{\Delta TC}{\Delta Q} = \frac{C_2 - C_1}{Q_2 - Q_1} = \frac{2000 - 1000}{200 - 100}$$

$$\therefore MC = 100 \text{ Ans}$$

(12) Ecological Footprint:

- Ecological footprint is a metric that measures how much nature is required to support human needs.
- Ecological footprint measures human demands on nature.
- The ecological footprint can be calculated by following formula;

$$EF = \frac{D}{Y}$$

where,

D = Annual demand of a product

Y = Annual yield of that same product.

→ The advantages of ecological footprint are as follows:-

- (i) It is a single unit that allows disaggregation of indicators.
- (ii) It is useful as a means of communication policy.
- (iii) It may be top down or bottom up.
- (iv) It is widely applied to various activities.
- (v) It analyzes the sustainable development.

(8) Sustainable Development:

- Sustainable development is the development that meets the needs of present generation without compromising the ability of future generation to meet their needs.
- Sustainable development is a continuous process.
- Sustainable development is also known as long-term development.
- Sustainable development seeks a balance between environmental protection, economic growth and social inclusion.
- The 3 primary objectives of sustainable development are as follows:-

(i) Environmental Protection;

→ Environmental protection includes:

- Preserving natural resources
- Reducing pollutions and
- mitigating climate change to ensure healthy planet for future generation.

(ii) Economic Growth;

→ Economic growth includes;

- Promoting economic activities that create jobs
- improving living standards and so on.

(iii) Social Inclusion:

→ Social inclusion includes:

- ~~promot~~ ensuring equal opportunities,
- reducing poverty
- promoting human rights, education and health care for all.

(3) Ecological Limit:

- Ecological limit refers to the boundaries or thresholds within which ecosystems can function sustainably.
- Ecological limits are determined by the availability of resources, the capacity of ecosystems to absorb waste and the resilience of biodiversity.
- When human activities exceed the ecological limits through deforestation, overfishing, pollution, and excessive carbon emissions, it leads to environmental degradation, loss of biodiversity and climate change.
- A well-known framework related ecological limit is the Planetary Boundaries concept.
- Crossing the planetary boundaries can result in irreversible damage to Earth's ecosystems.

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Chapter-3

Enterprise Financing and Capital Investment

(a) Debt financing vs equity financing:

S.N.	Equity Financing	Debt Financing
(i)	Equity financing involves selling a portion of a company's equity in return for capital.	Debt financing involves borrowing money and paying it back with interest.
(ii)	Equity financing can be categorized by shares and stocks.	Debt financing can be categorized by loan, debentures, bonds etc.
(iii)	Equity financing can be kept permanently.	Debt financing must be repaid or refinanced.
(iv)	In equity financing, equity providers are aggressive.	In debt financing, debt providers are conservative.
(v)	Equity financing grants ownership of the firm.	Debt financing does not grant ownership of the firm.
(vi)	Equity financing falls under high-risk investments.	Debt financing falls under low-risk investments.
(vii)	Equity financing is an asset of the company.	Debt-financing is a loan of the company.